

- [5] Calculate the arithmetic mean and the median of the frequency distribution given below,

Height (in cms)	No. of students	Height (in cms)	No. of students
130 - 134	5	150 - 154	17
135 - 139	15	155 - 159	10
140 - 144	28	160 - 164	1
145 - 149	24		

Hence calculate the mode using empirical relation between the three.

Solution :

Height (in cms)	No. of, (f_i)	$f_i x_i$	cumulative
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The following table gives the distribution of weekly wages of 600 workers of a factory.

weekly wages (in Rs)	Frequency	Weekly wages (in Rs)	Frequency
Below - 375	69	600 - 675	58
375 - 450	167	675 - 750	24
450 - 525	207	750 - 825	10
525 - 600	65		

- Draw an ogive data and hence obtain the median value. Check it against calculated value.
- Obtain the limits of weekly wages of central 50 percent of the workers.

$$\sum m = 30$$

3

The ~~mean~~ median and mode of the following wage distribution are Rs. 33.5 and Rs. 34 respectively, however three frequencies are missing. Determine their values.

Wages	0-10	10-20	20-30	30-40	40-50	50-60	60-70	Total
frequencies	4	16	?	?	?	5	4	230

Solution:

Problem: 4

The following data relate to the sales of 100 companies.
Calculate mean, mode, median.

Sales (Rs. lakhs)	No. of companies	Sales (Rs. lakhs)	No. of companies
Below 60	12	66 - 68	10
60 - 62	18	68 - 70	3
62 - 64	25	70 - 72	2
64 - 66	30		N = 100

Solution:

Sales (Rs.)	No. of
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affected by extreme values.
Solved problem

Problem : 1 :

1. 1500 workers are working in an industrial establishment. Their age is classified as follows:

Age (Years)	No. of workers	Age (Years)	No. of workers
18 - 22	120	38 - 42	184
22 - 26	125	42 - 46	162
26 - 30	280	46 - 50	86
30 - 34	260	50 - 54	75
34 - 38	155	54 - 58	53

calculate the median age.

Solution :